Find, evaluate and select digital data sources

Digital Methods

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22nd of April 2025

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Who am I?

BA in European Ethnology

5th semester of the Social Data Science Master's

Internship at Will & Agency/PUBLIKUM, a Consultancy

Student assistant now at the consultancy working as a *Researcher & Data Scientist*

Any questions from the lecture?

Milestone 1 – The first Assignment

- **Milestone 1** – Friday, 25th of April, 23:59, via email to TA (hsl448@sodas.ku.dk)
- Feedback – next seminar
- Submission should contain:
  1. Description and justification of research topic and question
  2. Description and justification for central datasite(s)
- Remember to book your supervision slot for Thursday – seminar hours are held as usual.
- Starting next week one slide update **every Tuesday**

Scouting and evaluating datasites

It's important to know what to look for when scouting and choosing a datasite.

When selecting a datasite, you need to check if it is realistic to use the datasite, and ways to do this are presented by both Kozinets and Venturini.

Investigation process for datasite (Kozinets, 2020, p.215)

1. **Simplification**: Translate the research question into keywords/search terms
2. **Search operation**: Use search terms, hashtags, and trends to find potential datasites
3. **Scouting operations**: Observe and filter sites to narrow down choices
4. **Evaluating**: Apply evaluation criteria to choose datasites
5. **Selecting**: Capture and store relevant data (screenshots, copy/paste, scraping)

Quality Criteria for the datasites (Kozinets, 2020, pp.227-229)

- **Relevance:** Connection to research question (beyond just keywords)
- **Activity:** Recency and regularity of information/updates
- **Interactivity**: Two-or-more-way communication between participants
- **Diversity:** Expression of different perspectives (not single-voice dominated)
- **Richness:** Presence of detail, description, emotion and interconnection ("deep data")

Reality Checklist (Venturini et al., 2018)

- ✓ Consider the platform's role in your study object - *How much of your object of study occurs in the medium that you are studying?*
- ✓ Balance theoretical relevance with source-specific practices
- ✓ Account for data access limitations (APIs, webpages, sampling)
- ✓ Understand how data is "given" by the medium

Kozinets, R. V. (2020). *Netnography: The Essential Guide to Qualitative Social Media Research* (3rd ed.). SAGE Publications.
Venturini, T., Bounegru, L., Gray, J., & Rogers, R. (2018). A reality check(list) for digital methods. *New Media & Society, 20*(11), 4195-4217. https://doi.org/10.1177/1461444818769236

Ethical considerations and immersing

Always think of how you collect data and the ethical considerations.

Four immersive operations

(Kozinets, 2020, p.284-297)

1. **Reconnoitering:** Making reconnaissance of your data landscape
   - Providing picture of data landscape/mapping
   - Using "telescope and microscope" approaches (zooming in and out)
2. **Recording:** Documenting your data collection journey
   - Not just downloading data but documenting interpretations and reactions
3. **Researching**: Making sense of the data
   - Creating conceptual notes connecting observations to theory
   - Abstracting from concrete cases to theorize and formulate hypotheses
4. **Reflecting:** Documenting your reactions and perspective
   - Challenge your bias, habituated thinking, and affective reactions

Ethical considerations in selection

(Kozinets, 2020, pp.163-187)

- **Key Principles**: Autonomy, Beneficence, Non-maleficence, Confidentiality, Integrity
- **Site Classification**: Determine if sites are public or private
- **Risk Assessment**: Sensitive topics and vulnerable populations
- **Data Treatment:** Uncloaked, cloaked, or highly cloaked data
- **Ethics Flowchart**: Decision tree for ethical research procedures

Kozinets, R. V. (2020). *Netnography: The Essential Guide to Qualitative Social Media Research* (3rd ed.). SAGE Publications.

To do list

TUESDAY & THURSDAY

- 1) Scout for datasites
- 2) Evaluate datasites
- 3) Select one or more datasites to proceed with as the main datasource(s).
- 4) Describe relevant datasites in immersion journal
- 5) Update research question and design if needed.

Agenda

1. Evaluate data site

- First, evaluate the current data sites you are considering, using Kozinets's criterias: **relevance, activity, interactivity, diversity, and richness**.

2. Evaluate data traces

- Evaluate the **quality** of each central data trace on your data site – evaluate each of these both in relation to its **type** (i.e., textual, numerical, visual, etc.) and in relation to your **RQ**.

25 min

3. Describe the data generation process

- Describe how your data site **works**. How do you produce content? How do you consume content? How do you interact? How does one become a member?
- Describe **who** is on your data site. Who is active? Who can become active? Who controls the forum? Who owns your data site?

4. Data site selection

- Based on the steps above, **choose** your primary and secondary (if any) data sites.

45 min

5. Revisit your RQ

- Based on your work above, revisit your RQ. Are there any **adjustments** you need to make, different theoretical lenses to consider?

6. Work on your milestone

45 min

Possible ways of analysis

Ideas of how to analyze your data

Quantitative text analysis methods

- **Topic Modeling (LDA, NMF)**
  Automatically group text into topics based on word distribution patterns.
- **Keyword / Dictionary Analysis**
  Measure the frequency of pre-defined words or categories
- **Sentiment Analysis**
  Classify text by sentiment polarity (positive, negative, neutral) or emotional tone.
- **Frequency & N-gram Analysis**
  Count word, phrase (bigram, trigram) frequencies to detect trends and patterns.
- **Supervised Machine Learning**
  Classify texts (e.g., spam vs. non-spam, fake news vs. real) using labeled data.
- **Unsupervised Methods**
  Cluster texts based on similarity without pre-defined labels.
- **Temporal Analysis (Over Time)**
  Track changes in word use, sentiment, or topic prevalence over time.

Qualitative text analysis methods

- **Thematic Coding / Manual Annotation**
  Manually categorize content by themes or frames using qualitative coding schemes.
- **Semantic Network Analysis**
  Map relationships between words, concepts, or actors within text (who talks to/about whom and how).
- **Close Reading / Discourse Analysis**
  Deeply interpret selected texts to unpack narrative structures, rhetorical strategies, framing, or ideology.
- **Narrative / Storytelling Analysis**
  Examine how stories, arguments, or positions are constructed in text.